

Chapter 4

12. The suspension of a car can be modeled by the mass-spring-dashpot system of figure E.12. Let the mass m of the car be 1500 kg, let the spring constant K_s be 75000 N/m, and let the shock absorber (dashpot) viscosity coefficient K_d be 20000 N · s/m. At a certain length d_0 of the spring, it is unstretched and uncompressed and exerts no force. Let that length be 0.6 m.
- What is the distance $y(t) - x(t)$ when the car is at rest?
 - Define a new variable $z(t) = y(t) - x(t) - \text{constant}$ such that, when the system is at rest, $z(t) = 0$, and write a describing equation in z and x that describes an LTI system. Then find the impulse response.
 - The effect of the car striking a curb can be modeled by letting the road surface height change discontinuously by the height of the curb h_c . Let $h_c = 0.15$ m. Graph $z(t)$ versus time after the car strikes a curb.

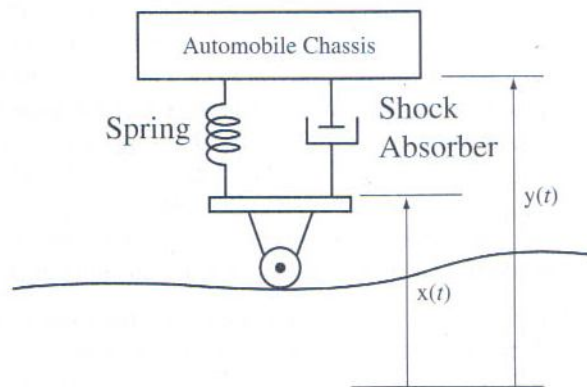


Figure E.12
Car suspension model.

20. A system is described by the equation $y(t) = \int_{-\infty}^{t/3} x(\lambda) d\lambda$. Classify the system as to time-invariance, stability and invertibility.

Chapter 6

18. Find the impulse responses of the systems described by these equations.

- $4y''(t) = 2x(t) - x'(t)$
- $y''(t) + 9y(t) = -6x'(t)$
- $-y''(t) + 3y'(t) = 3x(t) + 5x''(t)$

(c): (plot II-type block diagram with integrators)

22. Graph these functions.

- $g(t) = 3\cos(10\pi t) * 4\delta(t + 1/10)$
- $g(t) = \text{tri}(2t) * \delta_1(t)$

Chapter 15

33. Find the time-domain functions that are the inverse Laplace transforms of these functions. Then, using the initial- and final-value theorems, verify that they agree with the time-domain functions.

(a) $G(s) = \frac{4s}{(s+3)(s+8)}$

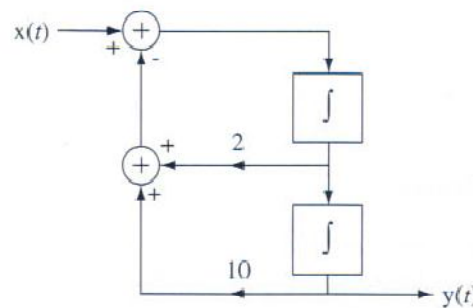
(b) $G(s) = \frac{4}{(s+3)(s+8)}$

(c) $G(s) = \frac{s}{s^2 + 2s + 2}$

(d) $G(s) = \frac{e^{-2s}}{s^2 + 2s + 2}$

39. Write the differential equations describing the systems in figure E.39 and find and graph the indicated responses.

(a) $x(t) = u(t)$, $y(t)$ is the response, $y(0^-) = -5$, $\left[\frac{d}{dt}(y(t)) \right]_{t=0^-} = 10$



(a): Use matlab to compute and plot the response of the system. You should use commands like 'tf', and 'step'.

43. Find the expression for the overall system transfer function of the system in figure E.43. For what positive values of K is the system stable?

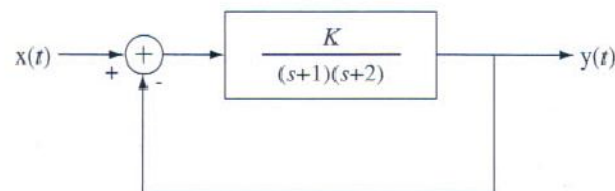


Figure E.43

For a fixed value of K that makes the system stable, use the matlab commands 'tf' and 'feedback' to compute the transfer function from x to y . Use 'impz' to plot the impulse response of the resulting system.